Conceptual question · Unit 4 · 2 points · 15 min · Conceptual / fast

C08 — Apsidal speeds and a perigee impulse: areal velocity, vis-viva and which apse moves

A satellite moves about the Earth (gravitational parameter μ, the Earth's centre F fixed at a focus) on a Kepler ellipse of eccentricity $e = 1/3$ and perigee radius r_p. Π is the perigee and A the apogee; a is the semi-major axis. Decide whether each statement is TRUE or FALSE and justify the answer. A verdict without a justification scores nothing.

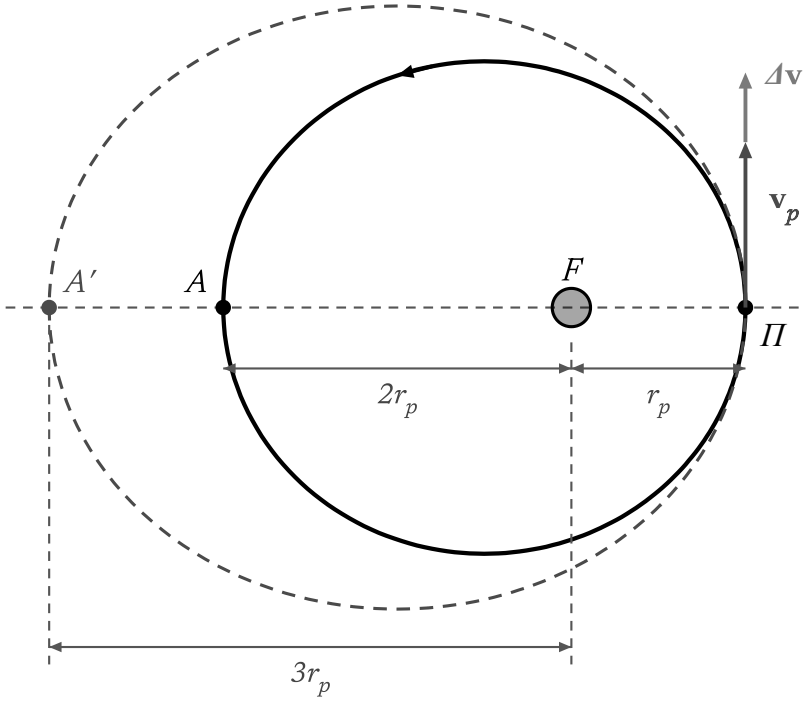


Figure 1. Orbit of eccentricity $e = 1/3$ about the focus F (perigee Π at r_p, apogee A at $r_a = 2r_p$) and, dashed, the orbit after a tangential impulse Δv at Π (apogee A').

Part (a) · 1 point

Speeds and areal velocity

Statement: "The speed at perigee is twice the speed at apogee; consequently the radius vector $\mathbf{r}$ from F to the satellite sweeps area twice as fast when the satellite passes the perigee as when it passes the apogee."

Step 1. Geometry of the ellipse

With $r_p = a(1-e)$ and $r_a = a(1+e)$: $a = \dfrac{r_p}{1-1/3} = \dfrac{3}{2}r_p$ and $r_a = \dfrac{3}{2}r_p \cdot \dfrac{4}{3} = 2r_p$.

Step 2. Speeds at the apsides

The force is central, so $\mathbf{h} = \mathbf{r} \times \mathbf{v}$ is constant (Notes §4.8.1). At both apsides $\mathbf{r} \perp \mathbf{v}$, hence $h = r_p v_p = r_a v_a$ and $\dfrac{v_p}{v_a} = \dfrac{r_a}{r_p} = 2$. The vis-viva equation $v^2 = \mu\left(\frac{2}{r} - \frac{1}{a}\right)$ (Notes §8.5) confirms it:

$$v_p^2 = \mu\left(\frac{2}{r_p} - \frac{2}{3r_p}\right) = \frac{4\mu}{3r_p}, \qquad v_a^2 = \mu\left(\frac{2}{2r_p} - \frac{2}{3r_p}\right) = \frac{\mu}{3r_p}, \qquad \frac{v_p^2}{v_a^2} = 4$$

Step 3. Areal velocity

The area swept per unit time is $\dot{A} = \frac{1}{2}|\mathbf{r} \times \mathbf{v}| = \frac{1}{2}h$, a constant: at perigee $\frac{1}{2} r_p v_p$, at apogee $\frac{1}{2}(2r_p)\,\frac{v_p}{2} = \frac{1}{2} r_p v_p$, the same. The satellite is faster near perigee precisely because it is closer, so that rv stays fixed. The conclusion is wrong: FALSE.

KEY POINTS

- FALSE (the first half is true).
- $r_a = r_p \dfrac{1+e}{1-e} = 2r_p$, $a = \frac{3}{2} r_p$; $h = r_p v_p = r_a v_a \Rightarrow v_p / v_a = 2$. Vis-viva: $v_p^2 = \dfrac{4\mu}{3r_p}$, which equals $\dfrac{2\mu}{a}$ only because $a = \frac{3}{2} r_p$ for this orbit; $v_a^2 = \dfrac{\mu}{3r_p}$.
- Areal velocity $\dot{A} = \frac{1}{2}|\mathbf{r} \times \mathbf{v}| = h/2$: the same at every point of the orbit (Kepler's second law).

Part (b) · 1 point

A tangential impulse at perigee

At Π *a short engine burn along the velocity raises the speed from* v_p *to* $v_p' = \sqrt{9/8}\, v_p$ *(an increase of about* $6.07\,\%$*) without changing the position. Statement: "The perigee radius is unchanged and the apogee radius grows from* $2r_p$ *to* $3r_p$*."*

Step 1. The burn point remains an apse

The impulse changes $\mathbf{v}$ along its own direction, so right after it $\mathbf{v}' \perp \mathbf{r}$ at $r = r_p$: the radial velocity is zero and Π is an apse of the new orbit. Since $v_p'^2 = \dfrac{3\mu}{2r_p}$ exceeds the circular value μ/r_p, the satellite is moving too fast to stay at r_p and rises: Π is the **perigee** of the new orbit.

Step 2. New energy and angular momentum

$$\varepsilon' = \frac{v_p'^2}{2} - \frac{\mu}{r_p} = \frac{3\mu}{4r_p} - \frac{\mu}{r_p} = -\frac{\mu}{4r_p} \;\Rightarrow\; a' = -\frac{\mu}{2\varepsilon'} = 2r_p, \qquad h'^2 = r_p^2 v_p'^2 = \tfrac{3}{2}\mu r_p$$

Step 3. Apse radii

At an apse $v = h'/r$, so $\varepsilon' = \dfrac{h'^2}{2r^2} - \dfrac{\mu}{r}$. Multiplying by r^2: $-\dfrac{\mu}{4r_p}r^2 + \mu r - \dfrac{3}{4}\mu r_p = 0$, i.e. $r^2 - 4r_p r + 3r_p^2 = (r - r_p)(r - 3r_p) = 0$. The perigee stays at r_p and the new apogee A' is at $3r_p$ (consistent with $r_p + r'_a = 2a' = 4r_p$), with $e' = \dfrac{3r_p - r_p}{3r_p + r_p} = \dfrac{1}{2}$. The statement is TRUE.

KEY POINTS

- TRUE.
- $v_p'^2 = \frac{9}{8} \cdot \dfrac{4\mu}{3r_p} = \dfrac{3\mu}{2r_p} > \dfrac{\mu}{r_p}$ (above circular speed, so Π stays the perigee).
- $\varepsilon' = -\dfrac{\mu}{4r_p}$, $a' = 2r_p$; $h'^2 = \frac{3}{2}\mu r_p$; the apse equation $\varepsilon' = \dfrac{h'^2}{2r^2} - \dfrac{\mu}{r}$ has roots $r = r_p$ and $r = 3r_p$; $e' = 1/2$.

Interpretation and checks

- **Dimensional check:** μ is $\mathrm{m^3\ s^{-2}}$, so μ/r is $\mathrm{m^2\ s^{-2}}$ (a speed squared and a specific energy); h^2/μ is a length.
- **Limits:** a burn that brings $v_p'^2$ to $2\mu/r_p$ gives $\varepsilon' = 0$ and an escape parabola ($r'_a \to \infty$); a retrograde burn to $v_p'^2 < \mu/r_p$ would turn Π into the new **apogee**.
- **Common mistakes in the exam:** (1) confusing a constant areal velocity with a constant speed (or the reverse); (2) thinking that a burn raises the point where it is applied, instead of the opposite apse; (3) forgetting that a tangential burn keeps the burn point an apse.